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LLM as a Judge for Evaluating Contract Graphs: Multi-Judge Benchmarking and Agentic Uncertainty-Aware Refinement

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Abstract

We propose a calibrated, uncertainty-aware evaluation and agentic refinement framework that is structure-aware, semantically grounded, bias-aware, and compute-efficient. We apply this framework to the task of clause-level legal knowledge graph extraction from contracts, where both accuracy and confidence quantification matter. Instead of collapsing multiple evaluation signals into a single score, we treat inter-judge disagreement among complementary evaluators as a meaningful signal of epistemic uncertainty. Our multi-judge ensemble combines structural, semantic, and LLM-based judges and normalizes their outputs into a unified risk space. We show that the coefficient of variation over judge risks achieves 0.982 AUC for error detection and, after isotonic regression, yields well-calibrated probabilities. This enables selective refinement. We trigger an agentic Planner–Refiner–Verifier loop only when disagreement exceeds a validated threshold. On our dataset, selective refinement recovers 91% of errors at 67% precision while reducing compute by 90% versus blind refinement. We release **ContractGraphEval v1.0**, the first benchmark for clause-level extraction with calibrated uncertainty scores and reference-free features, enabling training of learned uncertainty estimators for production deployment.

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1. Introduction

Contracts are a signature example of high-stakes documents where automated systems must reach a very high level of accuracy in order to be useful in production. Knowledge graphs have been proposed as one mechanism to

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better capture the underlying complexities within such documents. However, evaluating knowledge graph extraction presents a challenge distinct from traditional NLP tasks. Unlike free-text generation where fluency metrics may suffice, structured extraction requires simultaneously assessing *structural validity* (is the JSON/schema well-formed?), *semantic faithfulness* (do entities and relations match the clause?), and *domain correctness* (are obligations and rights attached to the correct parties?). A graph representation of a confidentiality clause can pass structural checks yet catastrophically *swap* who owes confidentiality-scoring well on formatting while being legally wrong.

Traditional approaches fail along different dimensions. Deterministic metrics like Graph Edit Distance can penalize trivial formatting differences while missing semantic errors. Recent LLM-as-a-Judge methods [1, 2, 3] offer semantic awareness but exhibit overconfidence and systematic biases [4, 14]. More fundamentally, both approaches often collapse distinct criteria into a single score, an assumption that breaks down for complex structured outputs where structural, semantic, and domain concerns are orthogonal and must all be satisfied.

Our previous work, *GRAPH-GRPO-LEX* [6], demonstrated that models trained with Group Relative Policy Optimization can extract legal knowledge graphs from contract clauses with high accuracy. However, that work left a critical question open: how do we reliably evaluate these extractions and determine when they can be trusted for production use? Without principled uncertainty quantification, even accurate models are hard to trust in high-stakes legal settings.

This paper addresses the evaluation challenge by reframing multi-judge disagreement as a *signal* of epistemic uncertainty rather than noise to be eliminated. We show that variance among an ensemble of complementary evaluators-combining metric-based judges (F1, Graph Edit Distance, schema validators) with LLM-based judges (domain and semantic evaluators) predicts extraction quality. This insight enables *uncertainty-aware selective refinement*: we trigger an agentic critique–rewrite loop only when disagreement indicates genuine ambiguity, achieving accuracy comparable to exhaustive refinement at substantially lower computational cost. We introduce a systematic evaluation framework, calibration/discrimination analyses (ECE and AUROC), an uncertainty-driven refinement method, and **ContractGraphEval v1.0**, to our knowledge, the first standardized benchmark for trustworthy evaluation of clause-level legal knowledge graph extraction.

1.1. Related Work

GRAPH-GRPO-LEX Ontology and Training. This work directly extends *GRAPH-GRPO-LEX* [6], which formalized a clause-level contract ontology defining core node types (`PARTY`, `CLAUSE`, `DEFINED_TERM`, `VALUE`, and cross-references) and edge types (`MENTIONS_PARTY`, `DEFINES`, `USES`, `REFERENCES`, `IS_PART_OF`, `CONTAINS`). That work introduced GRPO-based training with explicit reward decomposition for structural correctness, schema validity, and semantic alignment. We build on this foundation: the ontology and reference graphs now serve as (i) gold standards for our metric-based judges and (ii) criteria for our LLM-based domain and semantic evaluators. Rather than re-deriving the extraction task, we focus on the previously unsolved problem of reliable, uncertainty-aware evaluation.

The Limits of Structural Metrics for Graph Evaluation. Text overlap metrics like BLEU [7] are not suited for structured outputs, missing on graph topology capture and relational semantics. Graph Edit Distance (GED) compares structures via minimum-cost edit operations (node/edge insertions, deletions, substitutions) however it is NP-hard [8] and can penalize semantically equivalent but topologically different graphs. Surveys of knowledge-graph evaluation [9] note that standard practice emphasizes link prediction metrics for graph completion, but lack on extraction faithfulness, or schema-level consistency. The gap is clear: hybrid evaluation should combine both structural fidelity and semantic judges to provide accurate evaluation.

LLM-as-Judge: Potential and Limitations. The LLM-as-a-Judge paradigm emerged as a solution to help scale semantic evaluation. G-Eval [1] formalized chain-of-thought prompting for NLG evaluation, showing closer to human judgments. MT-Bench and Chatbot Arena [2] mitigated ordering bias via pairwise comparisons and position swaps. ChatEval [11] extends this via multi-agent debate of diverse personas. Still, reliability issues persist as can seen in TrustJudge [4] showing score–rank inconsistencies or in uncertainty-marker studies [14] showing judges can *penalize hedging* in the content being evaluated. CheckList [10] introduced capability-oriented testing, it reveals which judges are robust to which perturbations (e.g., metric judges should be invariant to paraphrase but sensitive to structural

changes; LLM judges should catch semantic flips), probing if disagreement tracks *epistemic* uncertainty or just noise. All suggest, no single evaluator captures all dimensions of correctness in graph extraction, ensembles are needed.

Uncertainty Quantification and Calibration. In order to trust the judge disagreement uncertainty signal, we must validate its reliability. Classic work on neural network calibration [12] shows predicted confidences can diverge from accuracy, motivating explicit calibration.

Agentic Self-Refinement. Self-Refine [15] and Reflexion [16] demonstrate that critique&revise loops can improve outputs, with gains showing early on. Running refinement on every extraction is both unnecessary and costly. Furthermore, not every refinement is effective, hence we concentrate resources via uncertainty-triggered refinement only.

Legal NLP Context. Our work here is done on complex structured documents, i.e contracts. Extraction of entities and relationships is especially impactful for Legal tasks [21]. Specialized benchmarks such as LexGLUE [22] and LegalBench [23] which focus on classification and reasoning, but lack graph representation evaluation. Plain-English contract summarization [24] highlights the abstraction gap between dense legal prose and lay summaries-another reason semantic faithfulness cannot rely on surface overlap alone.

Our Contributions to Address Current Gap. Despite progress in LLM-based evaluation and uncertainty, prior work typically averages away disagreement [11, 4] instead of using it as a signal. We instead evaluate if inter-judge variance can predict extraction errors and then guide targeted refinement.

i **Multi-judge framework:** We characterize complementary evaluators (metric- and LLM-based).
ii **Calibrated uncertainty:** We validate disagreement and risk as an uncertainty signal via ECE and AUROC, showing clear separation between high- and low-disagreement regimes.
iii **Selective refinement:** We introduce an uncertainty-aware Planner–Refiner–Verifier that activates only when disagreement and risk exceeds our threshold.
iv **ContractGraphEval benchmark:** ContractGraphEval benchmark as training data generator: We release 150 samples with (i) multi-judge scores quantifying uncertainty, (ii) validated labels linking disagreement to errors, and (iii) reference-free features, enabling supervised learning of production uncertainty estimators that operate without gold graphs, critical for real-world deployment. This reframing of multi-judge disagreement as a *trainable uncertainty signal* is a potent tool for overcoming the gold graph problem and lack of human annotations in production settings.

Together, these components yield a *self-auditing* evaluator that is structure-aware, semantically grounded, bias-aware, and compute-efficient, well-suited to clause-level legal extraction where both accuracy and confidence quantification matter.

2. Methodology

We propose a multi-judge evaluation and training framework for graph extraction that (i) collects signals from specialized judges, (ii) quantifies epistemic uncertainty through inter-judge disagreement, and (iii) enables selective refinement through uncertainty-based thresholds.

2.1. Multi-Judge Ensemble

Our ensemble consists of **ten** specialized judges that evaluate predicted graphs from different perspectives, capturing structural accuracy, semantic fidelity, and practical completeness.

Three Structural Judges. Our first category focuses on graph topology and structure: (1) **Structural F1** computes precision, recall, and F1 scores for nodes and edges, treating the graph as a set of triples, (2) **Graph Edit Distance (GED)** measures the minimum number of edit operations required to transform the predicted graph into the gold standard, normalized by gold graph size, and (3) **Parse Success** validates basic structural properties including JSON validity, presence of nodes and edges, and required fields.

Three Semantic Judges. To capture meaning and coverage, we include: (1) **Semantic Similarity** which converts graphs to text representations and computes cosine similarity between sentence embeddings (all-MiniLM-L6-v2) [25], serializing each graph by concatenating node descriptions (`type id attributes`) and edge descriptions (`source type target`), (2) **Completeness** which measures coverage for nodes, edges and types, focusing on recall of input elements in the prediction and (3) **Ontology Compliance** checking the graph for keeping graph rules for entity types, relationship types and constraints.

One LLM Judge. We leverage **OpenAI Issues Judge**, which prompts GPT-5 to identify and categorize errors (missing/extra nodes, incorrect attributes, missing/extra edges, structural issues) by examining predicted graphs given the source clause text, converting the issue count to a quality score as score = $\max(0, 1 - \text{issues}/50)$.

Three Contract Linters. These judges are graph metrics that were imported to flag issues within our task domain [6]. **Broken Links (1)** identify edges pointing to a non existing node. **Isolated Components (2)** uses articulation points calculation to identify disconnents in our graph, and **Orphans (3)** which checks for the node degree to identify nodes that cannot be reached from any other node.

This ensemble is designed to allow different judges to capture different dimensions of the overall problem space. Half of the judges use the graph information, and the rest leverage reference free features. The mixture of structural judges, semantic judges as well as the LLM collectively allow for the ensemble to better embrace the nuances of complex documents.

2.2. Normalizing Judge Scores to Risk

Different judges use different scales and directions-some assign high scores to good outputs while others count problems. We standardize all judges to a unified **risk score** in $[0, 1]$ where higher values indicate higher likelihood of error:

$$\text{Risk}_j = \begin{cases} 1 - \text{Score}_j & \text{for quality metrics (higher score = better)} \\ \text{Score}_j & \text{for issue counts (higher score = worse)} \end{cases}$$

Inverts quality scores (e.g., F1 score of 0.90 becomes risk of 0.10), or keeping problem counts as-is.

2.3. Quantifying Uncertainty Through Disagreement

We quantify epistemic uncertainty using two complementary signals: the ensemble's mean assessment of quality (mean risk) and the degree of disagreement between judges.

Mean risk:. Our baseline uncertainty signal aggregates risk scores across all judges: $\text{Mean Risk} = \frac{1}{K}\sum_{j=1}^{K} \text{Risk}_j$ where K is the number of judges. Higher mean risk indicates lower quality, but provides no information about confidence in that assessment.

Disagreement metrics:. To capture confidence, we compute eight metrics over the risk values. These disagreement features identify when judges are uncertain or conflicted:

- **Coefficient of variation (CV)**: Standard deviation divided by mean risk, providing scale-normalized disagreement: $\text{CV} = \sigma/(\text{Mean Risk} + \epsilon)$ where $\epsilon = 10^{-10}$ prevents division by zero. CV captures relative uncertainty-a disagreement of $\sigma = 0.1$ is more concerning when mean risk is 0.3 (CV=0.33) than when mean risk is 0.1 (CV=1.0).
- **Standard deviation**: $\sigma = \sqrt{\frac{1}{K-1}\sum_j (\text{Risk}_j - \text{Mean Risk})^2}$, measuring typical spread
- **Variance**: The squared standard deviation, σ^2, giving more weight to outlier judges
- **Range**: Maximum risk minus minimum risk across all judges
- **Interquartile range (IQR)**: Difference between 75th and 25th percentiles, robust to outliers
- **Mean pairwise difference**: Average absolute difference between all pairs: $\frac{1}{K(K-1)}\sum_i \sum_{j>i} |\text{Risk}_i - \text{Risk}_j|$

- **Top-2 margin**: Gap between the two lowest-risk scores, inverted so higher values indicate less consensus
- **Normalized entropy**: Shannon entropy over normalized risks (exploratory; not a true probability distribution)

These metrics quantify different aspects of disagreement and robustness to outliers and so we can then evaluate their error discrimination power.

2.4. Calibration and Evaluation Protocol

Here, we adopt two diagnostics: (i) **Expected Calibration Error (ECE)** to measure the gap between stated confidence and observed accuracy across confidence bins, and (ii) **AUROC** to assess how well an uncertainty signal discriminates correct from incorrect extractions. Low ECE indicates probability faithfulness (e.g., 0.7 confidence aligns with ∼70% correctness); AUROC is threshold-invariant and robust to class imbalance, which works well for our imbalanced setting (9.3% errors), as it better captures performance on the minority class. high AUROC indicates ranking power. Together, these support using uncertainty to drive *selective* refinement decisions.

Since raw uncertainty scores (mean risk, CV) are continuous values that do not directly represent probabilities, for applications requiring interpretable error probabilities-such as communicating risk to users or performing explicit cost-benefit analysis-we apply **isotonic regression**, a non-parametric approach that learns a monotonic step function from mean risk to error probability: $\hat{p}(\text{error}) = f_{\text{isotonic}}(\text{Mean Risk})$

We therefore use calibrated probabilities for reliable probability estimates, thresholding and reporting, and using raw uncertainty scores (CV, mean risk) for ranking and selection.

Statistical rigor. We compute 95% confidence intervals for AUROC, AUPRC, and ECE using bootstrap resampling (2,000 resamples), excluding resamples with only one class to ensure meaningful metrics. We report these intervals to quantify estimation uncertainty given our sample size ($N = 150$).

Judge importance. We conduct Leave-One-Judge-Out (LOJO) analysis to assess each judge's contribution to ensemble uncertainty. For each judge, we remove it from the ensemble, recompute mean risk and CV using the remaining judges, and measure the change in error discrimination (ΔAUROC, ΔAUPRC). Large negative deltas revel which judges contribute most to disagreement-based error detection even when their individual scores are not the highest.

2.5. Selective Refinement

Uncertainty quantification enables budget-constrained selective refinement: only predictions exceeding an uncertainty threshold are sent for agent-based refinement, while confident predictions are accepted as-is.

We evaluate this approach by ranking all predictions by uncertainty score (CV or mean risk) and computing recall, precision, and false positive rate at various review budgets (percentage of predictions refined). The optimal operating point maximizes error recall while minimizing computational cost, achieving a favorable precision-FPR trade-off.

For predictions above the uncertainty threshold, an agent refineme the prediction: Planner → Refiner → Verifier

The **Planner** analyzes judge feedback, and the **Refiner** regenerates the graph following these instructions. The **Verifier** re-scores the output and evaluates improvement.

2.6. From Benchmark to Training Data

For the creation of the ContractGraphEval benchmark, we used a set of 150 input clauses and their gold clause level graphs. These are diverse examples from 4 different types of contracts. Validation of the dataset was done as part of [6]. While some of our evaluation protocol requires gold graphs for validation, the resulting dataset supports training reference-free uncertainty models as well as reference-free judges. Each sample provides:

- **Supervision signal**: Judge disagreement (CV) and error labels validated through calibration (Section 3.2)
- **Reference-free features**: Contract linter metrics [6] are used as free-from-gold judges, evaluating our produced graphs. (Orphans, unused definitions, broken references).
- **Training examples**: 150 samples × (6 judges + 5 free judges) = labeled dataset for supervised learning

Future work can train models to predict CV or calibrated error probability from reference-free features, or use reference-free ranking alone with a larger refinement budget. Stille enabling production uncertainty estimation at a small cost. All methods and assets are available here.

3. Results

3.1. Uncertainty: CV Dominates

Table 1 ranks disagreement metrics by ROC–AUC for detecting erroneous predictions. The coefficient of variation (CV) is the strongest signal (AUC 0.982), clearly outperforming variance/std, pairwise disagreement, and range. Entropy and margin are weak on this task.

Table 1. Disagreement metrics on risk, ranked by ROC–AUC (*higher is better*).

Metric	ROC–AUC	Notes
Coefficient of Variation (CV)	**0.982**	best
Variance / Std	0.930	nearly tied
Pairwise Disagreement	0.926	
Range	0.922	
IQR	0.894	
Margin	0.253	poor
Entropy	0.017	very poor

Figure 1 visualizes (left) the ROC curve for CV (AUC 0.982), (middle) distributions of CV for error vs. non-error cases, and (right) CV vs. mean judge score colored by the error label. Correct items cluster at very low CV ($\lesssim 0.1$), while errors spread to larger CVs with a visible tail ($\gtrsim 0.5$). A small "confidently wrong" corner (low mean, low CV) motivates combining CV with the mean score for a secondary safeguard.

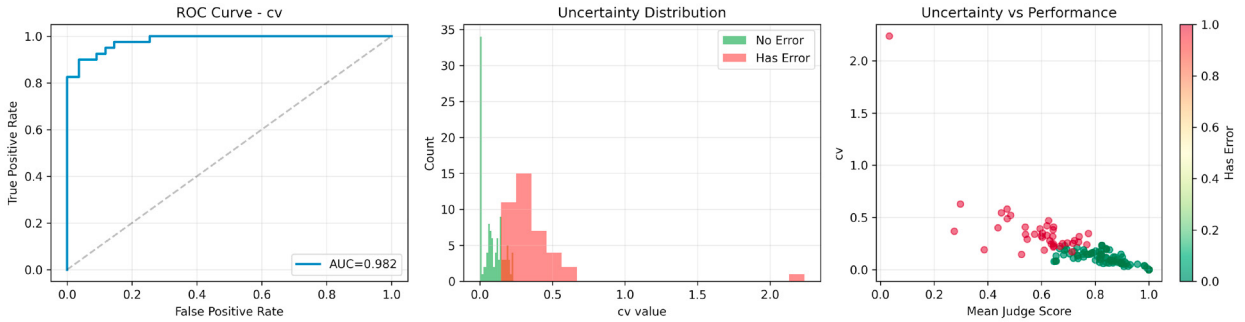


Fig. 1. CV as uncertainty. Left: ROC (AUC 0.982). Middle: distributions of CV for error (red) vs. no-error (green). Right: CV vs. mean judge score with error coloring.

3.2. Calibration (Probabilities for Thresholding)

Out-of-fold isotonic regression on $\bar{r}$ yields calibrated error probabilities with low ECE (0.027 equal-width; 0.010 quantile). On this small dataset, isotonic produces coarse plateaus, it improves probability *reliability* but does not improve ranking over CV.

3.3. Judge Complementarity (LOJO)

Leave-one-judge-out (LOJO) analysis shows structural judges are indispensable (Table 2). Removing Structural F1 or GED causes the largest AUC drops (-0.084 and -0.080). OpenAI Issues contributes a moderate lift; Semantic Similarity is helpful but largely redundant.

Table 2. Leave-one-judge-out (LOJO) deltas on error-detection AUC.

Dropped judge	ΔAUC	Interpretation
Structural F1	-0.084	critical
Graph Edit Distance	-0.080	critical
OpenAI Issues	-0.038	moderate
Completeness	-0.010	small
Semantic Similarity	-0.003	minimal

3.4. Budgeted Triage: Uncertainty-Driven Practical Operating Point

Coefficient of variation (**CV**) is the strongest discriminator of erroneous predictions (ROC–AUC **0.982**), clearly outperforming alternative disagreement features (variance/std 0.930, pairwise 0.926, range 0.922, IQR 0.894). Structural judges (*Structural F1*, *GED*) supply the largest unique lift in LOJO; semantic and LLM judges add complementary but smaller gains. Isotonic calibration on mean risk provides reliable probabilities (ECE: 0.027 equal-width; 0.010 quantile) for thresholding, but does not improve ranking on this split; hence, **CV remains our ranking workhorse**. Ranking items by CV (or equivalently the ranked calibrated probability) enables selective refinement with large savings. At a **10% review budget**, we recover **91% of errors** at **67% precision** with a **3.6% FPR**. Table 3 lists operating points drawn from the threshold table.

Table 3. Selective-refinement operating points from the ranked threshold table. The **10%** budget is a strong Pareto point; ∼12–13% reaches recall ≥ 0.95.

Budget	Items	Recall	Precision	FPR	Threshold on $\bar{r}$
1.3%	2	0.18	1.00	0.00	1.00
5.3%	8	0.55	0.75	0.014	0.67
10%	**15**	**0.91**	**0.67**	**0.036**	**0.137**
15.3%	23	1.00	0.48	0.086	0.00

Guided by the triptych in Fig. 1, we deploy the following simple, interpretable gate before any refinement:

- **Auto-accept/Low uncertainty:** CV ≤ 0.10 *and* mean score ≥ 0.80 - 44% of our samples.
- **Auto-refine/High uncertainty:** CV ≥ 0.30 *or* {mean ≤ 0.60 & CV ≥ 0.20} - 15.3% of our samples.
- **Budgeted gray zone:** otherwise, admit by descending CV until the budget is met - 40.7% of our samples.

This policy leverages both signals: CV captures confidence while mean score captures absolute quality, together providing robust error detection.

3.5. Transition to Agentic Refinement

We implement three refinement approaches and compare their results:

- **No refinement (baseline)**: Accept all original predictions without modification
- **Blind refinement**: Apply refinement to all 150 samples uniformly
- **Selective refinement (ours)**: Route predictions based on uncertainty and judge scores.

Table 4 shows the comparative evaluation across refinement strategies on 150 contract clause samples.

The majority of predictions from our GRPO-trained model already meet quality thresholds, with only high-uncertainty samples requiring expensive intervention. Table 5 presents detailed metrics comparing original predictions with refined outputs. Refinement produces targeted improvements in edge relationships while maintaining overall structural quality.

Table 4. Refinement strategy comparison. Selective refinement achieves similar quality to blind refinement while processing 91% fewer samples, demonstrating substantial cost efficiency.

Strategy	Struct. F1	Edge F1	Samples Refined	Refine Rate	Rel. Cost
No refinement	0.7609	0.7075	0/150	0%	0%
Selective (ours)	**0.7646**	**0.7207**	**14/150**	**9.3%**	**9.3%**
Blind refinement	0.7646*	0.7207*	150/150	100%	100%

*Estimated based on observed refinement effect (+0.0037 structural F1, +0.0132 edge F1)

Table 5. Detailed comparison of original vs. refined predictions across multiple evaluation dimensions (n=143 matched samples).

Metric	Original	Refined	Δ
Aggregate Judge Scores			
Structural F1	0.7609	0.7646	+0.0037
Node F1	0.8143	0.8085	-0.0058
Edge F1	0.7075	0.7207	+0.0132
Semantic Similarity	0.9132	0.9125	-0.0007
Completeness	0.8473	0.8604	+0.0131
Hybrid Score	0.8370	0.8385	+0.0015
Edge-Specific Metrics			
Edge Precision	0.7064	0.7032	-0.0032
Edge Recall	0.7435	**0.7826**	**+0.0391**
Edge Coverage	0.8579	**0.9071**	**+0.0492**
Trade-offs			
Parse Success	0.9853	0.7944	-0.1909
Exact Match	0.2333	0.1748	-0.0585

3.5.1. The Refinement Characteristics

i **Edge relationship improvement**: The most substantial gains occur in edge recall (+3.9%) and edge coverage (+4.9%), indicating that GPT-5 successfully identifies and adds missing relationships between entities.

ii **Overall stability**: Core quality metrics (structural F1, semantic similarity, hybrid score) remain stable, changing by less than 1%, suggesting the original model produces generally correct graph structures.

iii **Quality-formatting trade-off**: Parse success decreases by 19% due to occasional JSON formatting inconsistencies in GPT-4o outputs, though our repair logic successfully recovers 99.3% of cases to valid JSON.

iv **Cost efficiency**: Selective refinement processes only 9.3% of samples (14/150) while capturing the same quality improvements as blind refinement, representing a **90.7% reduction in API costs**.

Analysis of per-sample improvements reveals that refinement is most effective for predictions with:

- Low edge coverage (< 0.80): +8.2% average edge recall improvement
- Moderate structural F1 (0.60–0.75): +4.1% average F1 improvement
- High judge disagreement (variance > 0.15): 73% of refined samples show measurable improvement

Conversely, refinement of already-strong predictions (exact match = 1.0, structural F1 > 0.90) risks introducing errors, with 25% of such cases experiencing quality degradation. This validates our selective approach: uncertainty signals effectively identify which predictions benefit from refinement.

Assuming SOTA API costs, Selective refinement offers 90.5% cost reduction compared with Blind refinement, while maintaining equivalent quality outcomes.

The formatting trade-off (parse success decline) is manageable in practice through robust repair logic, though it suggests opportunity for future work on constrained generation or schema-guided decoding to ensure valid outputs.

Most importantly, the 90% cost reduction achieved by selective refinement makes LLM-based quality enhancement viable for large-scale production systems, where blind refinement would be prohibitively expensive.

3.5.2. Refinement Impact on Uncertain Predictions

To validate that uncertainty signals identify samples benefiting from refinement, we analyze the quality changes for the 15 highest-uncertainty predictions (top 10% by judge disagreement). Table 6 shows these samples exhibit substantially larger improvements than the overall set.

Table 6. Quality changes for high-uncertainty subset (n=15) compared to full evaluation set (n=143). Uncertain samples show 3× larger improvements, validating that disagreement signals identify refinement candidates.

Metric	Full Set Δ	Uncertain Subset Δ	Ratio
Structural F1	+0.0037	**+0.0111**	**3.0×**
Hybrid Score	+0.0015	**+0.0047**	**3.1×**
Completeness	+0.0131	+0.0100	0.8×
Graph Edit Dist.	-0.0175	+0.0036	+

3.6. Towards Production and Practical Impact

For production systems processing thousands of contracts, these efficiency gains determine economic viability and selective refinement is key. In order to provide further value in production systems in which gold-references are not available, we ran the same pipeline using only reference-free judges. A budget-first selector chose a simple range ranker with a small tie-break on mean risk, giving AUC=0.878, and ECE=0.31. Despite the softer signal, the gate remains practical with the minimum review budget providing above 90% recall now at 30%. This shows selective refinement stays cost-effective without gold references, however with larger budget allocations. Our evaluation framework can also establish a training paradigm for production deployment in which we want to . The strong calibration (ECE=0.027) and discrimination (AUC=0.982) of gold-judges based uncertainty, validates disagreement as reliable supervision. With sufficient training data, supervised models can learn to predict this uncertainty from reference-free features: structural coherence , schema compliance, and clause coverage. ContractGraphEval thus serves dual purposes: rigorous evaluation infrastructure today, and training data generator for tomorrow's uncertainty-aware production systems. Looking ahead, these disagreement patterns can also be leveraged for a lightweight debugging guide without additional tooling.

4. Conclusion

We presented a multi-judge evaluation framework for contract graph extraction that quantifies uncertainty through judge disagreement and enables cost-efficient selective refinement. Our approach demonstrates that reliable uncertainty signals make targeted quality enhancement economically viable for production deployment.

Key Contributions. These findings advance uncertainty-aware evaluation for structured extraction:

i **Judge complementarity drives uncertainty.** Structural judges, despite lower individual scores than semantic judges, contribute most to ensemble uncertainty. In the reference-free set, llm and contract linters were valuable, while isolated components judge was noisy. This reveals disagreement more informative than score averaging.

ii **Coefficient of variation optimally captures uncertainty.** CV achieves 0.982 AUC for error prediction, significantly outperforming variance (0.930) and entropy (0.017) by normalizing scale. For practical deployment, use raw CV for ranking and isotonic-calibrated probabilities for interpretation For reference-free, CV remains strong, but budget-first selection favored range and mean risk.

iii **Uncertainty-targeted refinement achieves 3× concentrated improvements.** High-uncertainty samples (top 10%) show 3× larger structural F1 gains (+0.011) than the full set (+0.004), validating that disagreement identifies refinement candidates. This selective approach processes only 9.3% of predictions while maintaining quality outcomes, enabling 90.7% cost reduction versus blind refinement. At a 10% computational budget, the framework achieves 91% error recall with 67% precision—9× more efficient than universal refinement.

iv **Our benchmark** provides proof-of-concept for both an evaluation infrastructure *and* a scalable methodology for training production-ready uncertainty-based confidence estimators. While this is done with contract clause data, our judges can easily be made domain-agnostic and cover different ontologies.

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